



Configuring Single-Area OSPF to Implement Dynamic Routing

Difficulty (out of 5 stars): ★★★

Recommended study time: 3 hours

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1 About This Lab

1.1 Task Description

An enterprise does not want to spend a large amount of manpower to maintain network performance, and requires that new routes be calculated based on route update packets when the network topology changes. In this lab, deploy OSPF to ensure network stability and interconnection.

After completing this task, you will gain a clear understanding of OSPF and be able to independently set up and configure a single-area OSPF network.

1.2 Objectives

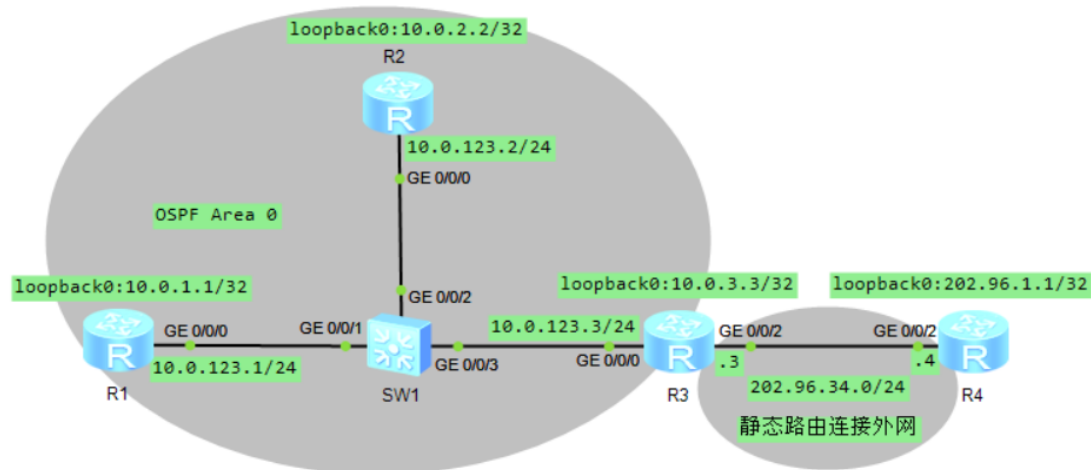
On completing this task, you will be able to:

1. Configure OSPF in a single area.
2. Configure OSPF area authentication.
3. Describe the process of establishing OSPF neighbor relationships on a multi-access network.
4. Change the cost of an OSPF interface.
5. Configure silent interfaces in OSPF.
6. Use display commands to check OSPF status.
7. Configure OSPF to advertise default routes.

2 Task Preparation

2.1 Network Topology

Figure 2-1 Lab topology





2.2 Initial Configuration

- Initial configuration of R1

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R1
[R1]int LoopBack 0
[R1-LoopBack0]ip address 10.0.1.1 32
[R1-LoopBack0]quit
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ip address 10.0.123.1 24
[R1-GigabitEthernet0/0/0]quit
[R1]
```

- Initial configuration of R2

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R2
[R2]int LoopBack 0
[R2-LoopBack0]ip address 10.0.2.2 32
[R2-LoopBack0]quit
[R2]int g0/0/0
[R2-GigabitEthernet0/0/0]ip address 10.0.123.2 24
[R2-GigabitEthernet0/0/0]quit
[R2]
```

- Initial configuration of R3

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R3
[R3]int lo0
[R3-LoopBack0]ip address 10.0.3.3 32
[R3-LoopBack0]quit
[R3]int g0/0/0
[R3-GigabitEthernet0/0/0]ip address 10.0.123.3 24
[R3-GigabitEthernet0/0/0]quit
[R3]int g0/0/2
[R3-GigabitEthernet0/0/2]ip address 202.96.34.3 24
[R3-GigabitEthernet0/0/2]quit
[R3]
```

- Initial configuration of R4. Note: R4 has a static route directed toward OSPF.

```
<Huawei>system-view
Enter system view, return user view with Ctrl+Z.
[Huawei]sysname R4
[R4]int LoopBack 0
[R4-LoopBack0]ip address 202.96.1.1 32
[R4-LoopBack0]quit
[R4]int g0/0/2
[R4-GigabitEthernet0/0/2]ip address 202.96.34.4 24
```



```
[R4-GigabitEthernet0/0/2]quit
[R4]
[R4]ip route-static 10.0.0.0 255.255.0.0 202.96.34.3
[R4]
```

3 Task Implementation

3.1 Checking Basic Configurations

Run the **display ip interface brief** command on R1, R2, R3, and R4 to check interface configurations.

```
<R1>display ip interface brief
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```

```
(s): spoofing
```

```
The number of interface that is UP in Physical is 3
```

```
The number of interface that is DOWN in Physical is 2
```

```
The number of interface that is UP in Protocol is 3
```

```
The number of interface that is DOWN in Protocol is 2
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.123.1/24	up	up
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	unassigned	down	down
LoopBack0	10.0.1.1/32	up	up(s)
NULL0	unassigned	up	up(s)

```
<R1>
```

```
<R2>display ip interface brief
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```

```
(s): spoofing
```

```
The number of interface that is UP in Physical is 3
```

```
The number of interface that is DOWN in Physical is 2
```

```
The number of interface that is UP in Protocol is 3
```

```
The number of interface that is DOWN in Protocol is 2
```

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.123.2/24	up	up
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	unassigned	down	down
LoopBack0	10.0.2.2/32	up	up(s)
NULL0	unassigned	up	up(s)

```
<R2>
```

```
[R3]display ip interface brief
```

```
*down: administratively down
```

```
^down: standby
```

```
(l): loopback
```



(s): spoofing

The number of interface that is UP in Physical is 4

The number of interface that is DOWN in Physical is 1

The number of interface that is UP in Protocol is 4

The number of interface that is DOWN in Protocol is 1

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	10.0.123.3/24	up	up
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	202.96.34.3/24	up	up
LoopBack0	10.0.3.3/32	up	up(s)
NULL0	unassigned	up	up(s)

[R4]display ip interface brief

*down: administratively down

^down: standby

(l): loopback

(s): spoofing

The number of interface that is UP in Physical is 3

The number of interface that is DOWN in Physical is 2

The number of interface that is UP in Protocol is 3

The number of interface that is DOWN in Protocol is 2

Interface	IP Address/Mask	Physical	Protocol
GigabitEthernet0/0/0	unassigned	down	down
GigabitEthernet0/0/1	unassigned	down	down
GigabitEthernet0/0/2	202.96.34.4/24	up	up
LoopBack0	202.96.1.1/32	up	up(s)
NULL0	unassigned	up	up(s)

[R4]

Run the **ping** command on R3 to check the connectivity between R3 and other devices.

[R3]ping 10.0.123.1

PING 10.0.123.1: 56 data bytes, press CTRL_C to break

Reply from 10.0.123.1: bytes=56 Sequence=1 ttl=255 time=60 ms

Reply from 10.0.123.1: bytes=56 Sequence=2 ttl=255 time=40 ms

Reply from 10.0.123.1: bytes=56 Sequence=3 ttl=255 time=50 ms

Reply from 10.0.123.1: bytes=56 Sequence=4 ttl=255 time=60 ms

Reply from 10.0.123.1: bytes=56 Sequence=5 ttl=255 time=40 ms

--- 10.0.123.1 ping statistics ---

5 packet(s) transmitted

5 packet(s) received

0.00% packet loss

round-trip min/avg/max = 40/50/60 ms

[R3]ping 10.0.123.2

PING 10.0.123.2: 56 data bytes, press CTRL_C to break

Reply from 10.0.123.2: bytes=56 Sequence=1 ttl=255 time=110 ms

Reply from 10.0.123.2: bytes=56 Sequence=2 ttl=255 time=30 ms



```
Reply from 10.0.123.2: bytes=56 Sequence=3 ttl=255 time=50 ms
Reply from 10.0.123.2: bytes=56 Sequence=4 ttl=255 time=50 ms
Reply from 10.0.123.2: bytes=56 Sequence=5 ttl=255 time=60 ms
```

```
--- 10.0.123.2 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 30/60/110 ms
```

[R3]ping 202.96.34.4

```
PING 202.96.34.4: 56 data bytes, press CTRL_C to break
Reply from 202.96.34.4: bytes=56 Sequence=1 ttl=255 time=30 ms
Reply from 202.96.34.4: bytes=56 Sequence=2 ttl=255 time=30 ms
Reply from 202.96.34.4: bytes=56 Sequence=3 ttl=255 time=20 ms
Reply from 202.96.34.4: bytes=56 Sequence=4 ttl=255 time=20 ms
Reply from 202.96.34.4: bytes=56 Sequence=5 ttl=255 time=20 ms
```

```
--- 202.96.34.4 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 20/24/30 ms
```

[R3]

Run the **ping** command to check the connectivity between R2 and other devices.

<R2>ping 202.96.23.3

```
PING 202.96.23.3: 56 data bytes, press CTRL_C to break
Reply from 202.96.23.3: bytes=56 Sequence=1 ttl=255 time=50 ms
Reply from 202.96.23.3: bytes=56 Sequence=2 ttl=255 time=20 ms
Reply from 202.96.23.3: bytes=56 Sequence=3 ttl=255 time=30 ms
Reply from 202.96.23.3: bytes=56 Sequence=4 ttl=255 time=20 ms
Reply from 202.96.23.3: bytes=56 Sequence=5 ttl=255 time=30 ms
```

```
--- 202.96.23.3 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 20/30/50 ms
```

<R2>

Run the **display ip routing-table** command to check the static routes configured on R4.

[R4]display ip routing-table

Route Flags: R - relay, D - download to fib

Routing Tables: Public

Destinations : 9 Routes : 9



Destination/Mask	Proto	Pre	Cost	Flags	NextHop	Interface
10.0.0.0/16	Static	60	0	RD	202.96.34.3	GigabitEthernet 0/0/2
127.0.0.0/8	Direct	0	0	D	127.0.0.1	InLoopBack0
127.0.0.1/32	Direct	0	0	D	127.0.0.1	InLoopBack0
127.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0
202.96.1.1/32	Direct	0	0	D	127.0.0.1	LoopBack0
202.96.34.0/24	Direct	0	0	D	202.96.34.4	GigabitEthernet 0/0/2
202.96.34.4/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/2
202.96.34.255/32	Direct	0	0	D	127.0.0.1	GigabitEthernet 0/0/2
255.255.255.255/32	Direct	0	0	D	127.0.0.1	InLoopBack0

[R4]

3.2 Configuring Single-Area OSPF

Configure the Loopback0 IP addresses of R1, R2, and R3 as their OSPF router IDs, and set the OSPF process ID to 1.

```
[R1]ospf 1 router-id 10.0.1.1
```

```
[R2]ospf 1 router-id 10.0.2.2
```

```
[R3]ospf 1 router-id 10.0.3.3
```

Activate OSPF on the interconnection and Loopback0 interfaces of R1, R2, and R3.

```
[R1]ospf 1
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0] network 10.0.123.1 0.0.0.0
[R1-ospf-1-area-0.0.0.0] network 10.0.1.1 0.0.0.0
```

```
[R2]ospf 1
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0] network 10.0.123.2 0.0.0.0
[R2-ospf-1-area-0.0.0.0] network 10.0.2.2 0.0.0.0
```

```
[R3]ospf 1
[R3-ospf-1]area 0
[R3-ospf-1-area-0.0.0.0] network 10.0.123.3 0.0.0.0
[R3-ospf-1-area-0.0.0.0] network 10.0.3.3 0.0.0.0
```

3.3 Setting the DR Priority of an Interface

The **ospf dr-priority** command sets a priority for an interface that participates in the DR election. A larger priority value indicates a higher priority. The value ranges from 0 to 255. The default value is 1. Configure R1 as the OSPF DR.

```
[R1]int g0/0/0
[R1-GigabitEthernet0/0/0]ospf dr-priority 10
```



3.4 Configuring OSPF Area Authentication

To ensure security, configure OSPF area authentication, use the cleartext mode, and set the password to **huawei**.

```
[R1]ospf 1
[R1-ospf-1]area 0
[R1-ospf-1-area-0.0.0.0] authentication-mode simple plain huawei

[R2]ospf 1
[R2-ospf-1]area 0
[R2-ospf-1-area-0.0.0.0] authentication-mode simple plain huawei

[R3]ospf 1
[R3-ospf-1]area 0
[R3-ospf-1-area-0.0.0.0] authentication-mode simple plain Huawei
```

3.5 Checking OSPF Neighbor Relationships

Check the OSPF neighbor information on R1, R2, and R3.

```
[R1]display ospf peer

OSPF Process 1 with Router ID 10.0.1.1
  Neighbors

Area 0.0.0.0 interface 10.0.123.1(GigabitEthernet0/0/0)'s neighbors
Router ID: 10.0.2.2      Address: 10.0.123.2
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 10.0.123.1  BDR: 10.0.123.3  MTU: 0
  Dead timer due in 33  sec
  Retrans timer interval: 5
  Neighbor is up for 00:02:18
  Authentication Sequence: [ 0 ]

Router ID: 10.0.3.3      Address: 10.0.123.3
  State: Full  Mode:Nbr is Master  Priority: 1
  DR: 10.0.123.1  BDR: 10.0.123.3  MTU: 0
  Dead timer due in 40  sec
  Retrans timer interval: 5
  Neighbor is up for 00:03:16
  Authentication Sequence: [ 0 ]

[R1]
```

The command output shows that R1 has established OSPF neighbor relationships with R2 and R3.

```
[R2]display ospf peer

OSPF Process 1 with Router ID 10.0.2.2
  Neighbors
```




```
Area 0.0.0.0 interface 10.0.123.2(GigabitEthernet0/0/0)'s neighbors
```

```
Router ID: 10.0.1.1      Address: 10.0.123.1
```

```
State: Full Mode:Nbr is Slave Priority: 1
```

```
DR: 10.0.123.1 BDR: 10.0.123.3 MTU: 0
```

```
Dead timer due in 35 sec
```

```
Retrans timer interval: 0
```

```
Neighbor is up for 00:02:52
```

```
Authentication Sequence: [ 0 ]
```

```
Router ID: 10.0.3.3      Address: 10.0.123.3
```

```
State: Full Mode:Nbr is Master Priority: 1
```

```
DR: 10.0.123.1 BDR: 10.0.123.3 MTU: 0
```

```
Dead timer due in 29 sec
```

```
Retrans timer interval: 5
```

```
Neighbor is up for 00:02:52
```

```
Authentication Sequence: [ 0 ]
```

[R2]

The command output shows that R2 has established OSPF neighbor relationships with R1 and R3.

[R3]display ospf peer

```
OSPF Process 1 with Router ID 10.0.3.3
```

```
Neighbors
```

```
Area 0.0.0.0 interface 10.0.123.3(GigabitEthernet0/0/0)'s neighbors
```

```
Router ID: 10.0.1.1      Address: 10.0.123.1
```

```
State: Full Mode:Nbr is Slave Priority: 1
```

```
DR: 10.0.123.1 BDR: 10.0.123.3 MTU: 0
```

```
Dead timer due in 38 sec
```

```
Retrans timer interval: 0
```

```
Neighbor is up for 00:04:33
```

```
Authentication Sequence: [ 0 ]
```

```
Router ID: 10.0.2.2      Address: 10.0.123.2
```

```
State: Full Mode:Nbr is Slave Priority: 1
```

```
DR: 10.0.123.1 BDR: 10.0.123.3 MTU: 0
```

```
Dead timer due in 36 sec
```

```
Retrans timer interval: 5
```

```
Neighbor is up for 00:03:35
```

```
Authentication Sequence: [ 0 ]
```

[R3]

The command output shows that R3 has established OSPF neighbor relationships with R1 and R2.

3.6 Checking OSPF Routing Tables

Check the OSPF neighbor routing tables of R1, R2, and R3.



```
[R1]display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.1.1
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.1.1/32	0	Stub	10.0.1.1	10.0.1.1	0.0.0.0
10.0.123.0/24	1	Transit	10.0.123.1	10.0.1.1	0.0.0.0
10.0.2.2/32	1	Stub	10.0.123.2	10.0.2.2	0.0.0.0
10.0.3.3/32	1	Stub	10.0.123.3	10.0.3.3	0.0.0.0

```
Total Nets: 4
```

```
Intra Area: 4  Inter Area: 0  ASE: 0  NSSA: 0
```

```
[R1]
```

The command output shows that R1 has learned the Loopback0 routes from R2 and R3.

```
[R2]display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.2.2
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.2.2/32	0	Stub	10.0.2.2	10.0.2.2	0.0.0.0
10.0.123.0/24	1	Transit	10.0.123.2	10.0.2.2	0.0.0.0
10.0.1.1/32	1	Stub	10.0.123.1	10.0.1.1	0.0.0.0
10.0.3.3/32	1	Stub	10.0.123.3	10.0.3.3	0.0.0.0

```
Total Nets: 4
```

```
Intra Area: 4  Inter Area: 0  ASE: 0  NSSA: 0
```

```
[R2]
```

The command output shows that R2 has learned the Loopback0 routes from R1 and R3.

```
[R3]display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.3.3
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.3.3/32	0	Stub	10.0.3.3	10.0.3.3	0.0.0.0
10.0.123.0/24	1	Transit	10.0.123.3	10.0.3.3	0.0.0.0
10.0.1.1/32	1	Stub	10.0.123.1	10.0.1.1	0.0.0.0
10.0.2.2/32	1	Stub	10.0.123.2	10.0.2.2	0.0.0.0

```
Total Nets: 4
```

```
Intra Area: 4  Inter Area: 0  ASE: 0  NSSA: 0
```

```
[R3]
```



The command output shows that R3 has learned the Loopback0 routes from R1 and R2.

3.7 Testing the Connectivity Between Loopback Interfaces

On R1, use the IP address of Loopback0 as the source address to ping the Loopback0 interfaces of R2 and R3.

```
[R1]ping -c 2 -a 10.0.1.1 10.0.2.2
PING 10.0.2.2: 56 data bytes, press CTRL_C to break
  Reply from 10.0.2.2: bytes=56 Sequence=1 ttl=255 time=80 ms
  Reply from 10.0.2.2: bytes=56 Sequence=2 ttl=255 time=40 ms

--- 10.0.2.2 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 40/60/80 ms

[R1]ping -c 2 -a 10.0.1.1 10.0.3.3
PING 10.0.3.3: 56 data bytes, press CTRL_C to break
  Reply from 10.0.3.3: bytes=56 Sequence=1 ttl=255 time=60 ms
  Reply from 10.0.3.3: bytes=56 Sequence=2 ttl=255 time=50 ms

--- 10.0.3.3 ping statistics ---
  2 packet(s) transmitted
  2 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 50/55/60 ms

[R1]
```

3.8 Checking the OSPF LSDB on R1

```
[R1]display ospf lsdb

OSPF Process 1 with Router ID 10.0.1.1
Link State Database

Area: 0.0.0.0
Type      LinkState ID      AdvRouter      Age  Len  Sequence      Metric
Router    10.0.3.3          10.0.3.3       685  48   80000006      0
Router    10.0.2.2          10.0.2.2       684  48   80000005      0
Router    10.0.1.1          10.0.1.1       684  48   80000008      0
Network   10.0.123.1        10.0.1.1       684  36   80000004      0

[R1]
```

The command output shows four LSAs, among which the first three are Type 1 LSAs generated by R1, R2, and R3 separately. The **AdvRouter** field for each LSA indicates the router that generates the LSA. The fourth LSA is a Type 2 LSA, which is generated by the DR on a network segment. In this scenario, R1 is the DR on the network segment 10.0.123.0/24. Therefore, the **AdvRouter** field value of the Type 2 LSA is 10.0.1.1.



3.9 Checking the Type 1 LSA Generated by R1

```
[R1]display ospf lsdb router self-originate
```

```
OSPF Process 1 with Router ID 10.0.1.1
```

```
Area: 0.0.0.0
```

```
Link State Database
```

```
Type      : Router
Ls id     : 10.0.1.1
Adv rtr   : 10.0.1.1
Ls age    : 794
Len       : 48
Options   : E
seq#      : 80000008
chksum    : 0x8f7b
Link count: 2
* Link ID: 10.0.1.1
  Data    : 255.255.255.255
  Link Type: StubNet
  Metric : 0
  Priority : Medium
* Link ID: 10.0.123.1
  Data    : 10.0.123.1
  Link Type: TransNet
  Metric : 1
```

```
[R1]
```

The command output shows that this LSA describes two Links. The first Link indicates the network segment where the loopback interface resides. The value of **Link Type** is **StubNet**, and the values of **Link ID** and **Data** are the IP address and mask of the stub network segment, respectively. The second link describes the network segment on which the three routers are interconnected. The value of **Link Type** is **TransNet**. The value of **Link ID** is the DR's interface address 10.0.123.1, and the value of **Data** is the IP address of the local interface on the network segment, 10.0.123.1.

3.10 Checking the Type 2 LSA Generated by R1

```
[R1]display ospf lsdb network self-originate
```

```
OSPF Process 1 with Router ID 10.0.1.1
```

```
Area: 0.0.0.0
```

```
Link State Database
```

```
Type      : Network
Ls id     : 10.0.123.1
Adv rtr   : 10.0.1.1
Ls age    : 897
Len       : 36
Options   : E
seq#      : 80000004
```



```
chksum      : 0x3f57
Net mask    : 255.255.255.0
Priority     : Low
Attached Router 10.0.1.1
Attached Router 10.0.2.2
Attached Router 10.0.3.3
```

[R1]

The **Attached Router** field in the Type 2 LSA describes the neighbor information of the network segment where the DR resides.

3.11 Checking the Process of Establishing OSPF Neighbor Relationships

Based on the preceding OSPF neighbor information, the interface IP address of the DR is 10.0.123.1, which is different from the expected result obtained based on DR election rules. In OSPF, the non-preemption mode is used for DR election. When a DR or BDR exists on a network, a newly deployed router on the network cannot preempt the role of the DR or BDR. However, if OSPF is configured at different time on the network devices, the elected DR may be the device that starts first.

You can shut down the interfaces connecting R1, R2, and R3, and run the **debugging ospf 1 event** command to observe the OSPF neighbor relationship establishment process. Then, try to re-enable the interfaces of R1, R2, and R3 simultaneously, and check the DR and BDR election process based on the debugging information output.

Shut down the interconnection interfaces of R1, R2, and R3.

```
[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/3] shutdown

[R2] interface GigabitEthernet0/0/0
[R2-GigabitEthernet0/0/4] shutdown

[R3] interface GigabitEthernet0/0/0
[R3-GigabitEthernet0/0/4] shutdown
```

Enable the debugging function on R1, R2, and R3, and enable OSPF event debugging.

```
[R1]quit
<R1>terminal debugging
Info: Current terminal debugging is on.
<R1>
<R1>terminal monitor
Info: Current terminal monitor is on.
<R1>
<R1>debugging ospf 1 event
<R1>

[R2]quit
<R2>terminal debugging
Info: Current terminal debugging is on.
<R2>
```



```
<R2>terminal monitor
Info: Current terminal monitor is on.
<R2>
<R2>debugging ospf 1 event
<R2>

[R3]quit
<R3>terminal debugging
Info: Current terminal debugging is on.
<R3>
<R3>terminal monitor
Info: Current terminal monitor is on.
<R3>
<R3>debugging ospf 1 event
<R3>
```

Re-enable the interconnection interfaces of R1, R2, and R3.

```
[R1] interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0] undo shutdown

[R2] interface GigabitEthernet0/0/0
[R2-GigabitEthernet0/0/0] undo shutdown

[R3] interface GigabitEthernet0/0/0
[R3-GigabitEthernet0/0/0] undo shutdown
```

Observe the debugging information on R3.

```
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:14-08:00 R3 %%01IFPDT/4/IF_STATE(l)[6]:Interface GigabitEthernet0/0/0 has turned into UP state.
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:14-08:00 R3 %%01IFNET/4/LINK_STATE(l)[7]:The line protocol IP on the interface GigabitEthernet0/0/0 has entered the UP state.
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:14.202.4-08:00 R3 RM/6/RMDEBUG:
  FileID: 0xd017802c Line: 1295 Level: 0x20

  OSPF 1: Intf 10.0.123.3 Rcv InterfaceUp State Down -> Waiting.
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:14.202.5-08:00 R3 RM/6/RMDEBUG:
  FileID: 0xd017802c Line: 1409 Level: 0x20

  OSPF 1 Send Hello Interface Up on 10.0.123.3
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:47-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(l)[8]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=2.123.0.10, NeighborEvent=HelloReceived, NeighborPreviousState=Down, NeighborCurrentState=Init)
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:47.602.2-08:00 R3 RM/6/RMDEBUG:
```



FileID: 0xd017802d Line: 1136 Level: 0x20

OSPF 1: Nbr 10.0.123.2 Rcv HelloReceived State Down -> Init.

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:50-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[9]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=1.123.0.10, NeighborEvent=HelloReceived, NeighborPreviousState=Down, NeighborCurrentState=Init)

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:50.552.2-08:00 R3 RM/6/RMDEBUG:

FileID: 0xd017802d Line: 1136 Level: 0x20

OSPF 1: Nbr 10.0.123.1 Rcv HelloReceived State Down -> Init.

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:51-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[10]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=1.123.0.10, NeighborEvent=2WayReceived, NeighborPreviousState=Init, NeighborCurrentState=2Way)

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:51-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[11]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=2.123.0.10, NeighborEvent=2WayReceived, NeighborPreviousState=Init, NeighborCurrentState=2Way)

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:51.462.3-08:00 R3 RM/6/RMDEBUG:

FileID: 0xd017802d Line: 1732 Level: 0x20

OSPF 1: Nbr 10.0.123.1 Rcv 2WayReceived State Init -> 2Way.

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:51.462.4-08:00 R3 RM/6/RMDEBUG:

FileID: 0xd017802d Line: 1732 Level: 0x20

OSPF 1: Nbr 10.0.123.2 Rcv 2WayReceived State Init -> 2Way.

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:56-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[12]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=1.123.0.10, NeighborEvent=AdjOk?, NeighborPreviousState=2Way, NeighborCurrentState=ExStart)

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:56-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[13]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=2.123.0.10, NeighborEvent=AdjOk?, NeighborPreviousState=2Way, NeighborCurrentState=ExStart)

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:57-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[14]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=1.123.0.10, NeighborEvent=NegotiationDone, NeighborPreviousState=ExStart, NeighborCurrentState=Exchange)

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:57-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[15]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=2.123.0.10, NeighborEvent=NegotiationDone, NeighborPreviousState=ExStart, NeighborCurrentState=Exchange)

[R3-GigabitEthernet0/0/0]

Apr 3 2025 21:23:57-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[16]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=1.123.0.10, NeighborEvent=NegotiationDone, NeighborPreviousState=ExStart, NeighborCurrentState=Exchange)



```
orEvent=ExchangeDone, NeighborPreviousState=Exchange, NeighborCurrentState=Loading)
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:57-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[17]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=1.123.0.10, NeighborEvent=LoadingDone, NeighborPreviousState=Loading, NeighborCurrentState=Full)
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:57-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[18]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=2.123.0.10, NeighborEvent=ExchangeDone, NeighborPreviousState=Exchange, NeighborCurrentState=Loading)
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:57-08:00 R3 %%01OSPF/4/NBR_CHANGE_E(I)[19]:Neighbor changes event: neighbor status changed. (ProcessId=256, NeighborAddress=2.123.0.10, NeighborEvent=LoadingDone, NeighborPreviousState=Loading, NeighborCurrentState=Full)
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:56.982.1-08:00 R3 RM/6/RMDEBUG:
  FileID: 0xd017802d Line: 1732 Level: 0x20

  OSPF 1: Nbr 10.0.123.1 Rcv AdjOk? State 2Way -> ExStart.
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:56.982.2-08:00 R3 RM/6/RMDEBUG:
  FileID: 0xd017802d Line: 1732 Level: 0x20

  OSPF 1: Nbr 10.0.123.2 Rcv AdjOk? State 2Way -> ExStart.
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:56.982.3-08:00 R3 RM/6/RMDEBUG:
  FileID: 0xd017802c Line: 2096 Level: 0x20

  OSPF 1 Send Hello Interface State Changed on 10.0.123.3
Apr  3 2025 21:23:56.982.4-08:00 R3 RM/6/RMDEBUG:
  FileID: 0xd017802c Line: 2107 Level: 0x20

  OSPF 1: Intf 10.0.123.3 Rcv WaitTimer State Waiting -> DROther.
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:57.22.1-08:00 R3 RM/6/RMDEBUG:
  FileID: 0xd017802d Line: 1845 Level: 0x20

  OSPF 1: Nbr 10.0.123.1 Rcv NegotiationDone State ExStart -> Exchange.
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:57.42.1-08:00 R3 RM/6/RMDEBUG:
  FileID: 0xd017802d Line: 1845 Level: 0x20

  OSPF 1: Nbr 10.0.123.2 Rcv NegotiationDone State ExStart -> Exchange.
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:57.72.1-08:00 R3 RM/6/RMDEBUG:
  FileID: 0xd017802d Line: 1957 Level: 0x20

  OSPF 1: Nbr 10.0.123.1 Rcv ExchangeDone State Exchange -> Loading.
[R3-GigabitEthernet0/0/0]
Apr  3 2025 21:23:57.72.2-08:00 R3 RM/6/RMDEBUG:
```




```
FileID: 0xd017802d Line: 2356 Level: 0x20
```

```
OSPF 1: Nbr 10.0.123.1 Rcv LoadingDone State Loading -> Full.
```

```
[R3-GigabitEthernet0/0/0]
```

```
Apr  3 2025 21:23:57.92.1-08:00 R3 RM/6/RMDEBUG:
```

```
FileID: 0xd017802d Line: 1957 Level: 0x20
```

```
OSPF 1: Nbr 10.0.123.2 Rcv ExchangeDone State Exchange -> Loading.
```

```
[R3-GigabitEthernet0/0/0]
```

```
Apr  3 2025 21:23:57.92.2-08:00 R3 RM/6/RMDEBUG:
```

```
FileID: 0xd017802d Line: 2356 Level: 0x20
```

```
OSPF 1: Nbr 10.0.123.2 Rcv LoadingDone State Loading -> Full.
```

```
[R3-GigabitEthernet0/0/0]
```

The preceding information shows the process of establishing an OSPF neighbor relationship.

3.12 Configuring Network Types of OSPF Interfaces

In the OSPF routing table on R1, check the Loopback0 routes learned from R2 and R3.

```
<R1>display ospf routing 10.0.2.2
```

```
OSPF Process 1 with Router ID 10.0.1.1
```

```
Destination : 10.0.2.2/32
```

```
AdverRouter : 10.0.2.2
```

```
Area       : 0.0.0.0
```

```
Cost       : 1
```

```
Type       : Stub
```

```
NextHop    : 10.0.123.2
```

```
Interface  : GigabitEthernet0/0/0
```

```
Priority    : Medium
```

```
Age        : 00h09m25s
```

```
<R1>
```

```
<R1>display ospf routing 10.0.3.3
```

```
OSPF Process 1 with Router ID 10.0.1.1
```

```
Destination : 10.0.3.3/32
```

```
AdverRouter : 10.0.3.3
```

```
Area       : 0.0.0.0
```

```
Cost       : 1
```

```
Type       : Stub
```

```
NextHop    : 10.0.123.3
```

```
Interface  : GigabitEthernet0/0/0
```

```
Priority    : Medium
```

```
Age        : 00h10m06s
```

```
<R1>
```

Take R2 as an example. Check OSPF Type 1 LSAs on R2.

```
<R2>display ospf lsdb router 10.0.2.2
```

```
OSPF Process 1 with Router ID 10.0.2.2
```

```
Area: 0.0.0.0
```

```
Link State Database
```

```
Type       : Router
```



```
Ls id      : 10.0.2.2
Adv rtr    : 10.0.2.2
Ls age     : 654
Len        : 48
Options    : E
seq#       : 8000000b
chksum     : 0x946c
Link count: 2
  * Link ID: 10.0.2.2
    Data    : 255.255.255.255
    Link Type: StubNet
    Metric  : 0
    Priority: Medium
  * Link ID: 10.0.123.1
    Data    : 10.0.123.2
    Link Type: TransNet
    Metric  : 1
```

<R2>

The command output shows that the mask of the LSA describing Loopback0 on R2 has been set to 32 bits. OSPF considers a loopback interface as a stub network, which is connected to only one node. Therefore, no matter how many bits are actually set for the network mask of the loopback interface, OSPF uses a 32-bit network mask (host mask) when advertising the Type 1 LSA to describe this interface.

3.13 Changing the Costs of OSPF Interfaces

In the OSPF routing table on R1, check the Loopback0 route learned from R3.

```
<R1>display ospf routing 10.0.3.3
```

```
OSPF Process 1 with Router ID 10.0.1.1
```

Destination : 10.0.3.3/32	
AdverRouter : 10.0.3.3	Area : 0.0.0.0
Cost : 1	Type : Stub
NextHop : 10.0.123.3	Interface : GigabitEthernet0/0/0
Priority : Medium	Age : 00h10m06s

```
<R1>display ospf routing 10.0.3.3
```

```
OSPF Process 1 with Router ID 10.0.1.1
```

Destination : 10.0.3.3/32	
AdverRouter : 10.0.3.3	Area : 0.0.0.0
Cost : 1	Type : Stub
NextHop : 10.0.123.3	Interface : GigabitEthernet0/0/0
Priority : Medium	Age : 00h13m52s

<R1>

<R1>

The command output shows that the route cost is 1.

Change the OSPF cost of GE0/0/0 on R1 to 20 and that of GE0/0/0 on R3 to 10.



```
[R1]interface GigabitEthernet0/0/0
[R1-GigabitEthernet0/0/0] ospf cost 20
```

```
[R3]interface GigabitEthernet0/0/0
[R3-GigabitEthernet0/0/0] ospf cost 10
```

In the OSPF routing table on R1, check the Loopback0 route learned from R2 again.

```
[R1]display ospf routing 10.0.2.2
```

OSPF Process 1 with Router ID 10.0.1.1

Destination :	10.0.2.2/32	Area	: 0.0.0.0
AdverRouter :	10.0.2.2	Type	: Stub
Cost	: 20	Interface :	GigabitEthernet0/0/0
NextHop	: 10.0.123.2	Age	: 00h00m57s
Priority	: Medium		

```
[R1]
```

The command output shows that the cost is 20.

In the OSPF routing table on R3, check the Loopback0 route learned from R1.

```
[R3]display ospf routing 10.0.1.1
```

OSPF Process 1 with Router ID 10.0.3.3

Destination :	10.0.1.1/32	Area	: 0.0.0.0
AdverRouter :	10.0.1.1	Type	: Stub
Cost	: 10	Interface :	GigabitEthernet0/0/0
NextHop	: 10.0.123.1	Age	: 00h01m38s
Priority	: Medium		

```
[R3]
```

The command output shows that the cost is 10.

3.14 Configuring an OSPF Silent Interface

Configure GE0/0/0 of R1 as a silent interface.

```
[R1]ospf 1
[R1-ospf-1] silent-interface GigabitEthernet 0/0/0
```

Check the OSPF neighbor information on R1.

```
<R1>display ospf peer
```

OSPF Process 1 with Router ID 10.0.1.1

After the configuration is completed, the interconnection interface no longer sends or receives Hello packets. The neighbor relationship that has been established using this interface disappears.

Check information about the OSPF interface GE0/0/0 on R1.

```
[R1-ospf-1]display ospf interface GigabitEthernet 0/0/0
```



```
OSPF Process 1 with Router ID 10.0.1.1
```

```
Interfaces
```

```
Interface: 10.0.123.1 (GigabitEthernet0/0/0)
```

```
Cost: 20      State: DR      Type: Broadcast      MTU: 1500
```

```
Priority: 1
```

```
Designated Router: 10.0.123.1
```

```
Backup Designated Router: 0.0.0.0
```

```
Timers: Hello 10 , Dead 40 , Poll 120 , Retransmit 5 , Transmit Delay 1
```

```
Silent interface, No hellos
```

```
[R1-ospf-1]
```

The command output shows that the interface is configured as a silent interface and no Hello packet exists on the interface.

Delete the silent interface configuration on R1.

```
[R1-ospf-1]
```

```
[R1-ospf-1]undo silent-interface GigabitEthernet 0/0/0
```

```
[R1-ospf-1]
```

Configure the Loopback0 interfaces on R2 and R3 as silent interfaces.

```
[R2]ospf 1
```

```
[R2-ospf-1] silent-interface LoopBack 0
```

```
[R3]ospf 1
```

```
[R3-ospf-1] silent-interface LoopBack 0
```

Check the OSPF routing table on R1.

```
[R1]display ospf routing
```

```
OSPF Process 1 with Router ID 10.0.1.1
```

```
Routing Tables
```

```
Routing for Network
```

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.1.1/32	0	Stub	10.0.1.1	10.0.1.1	0.0.0.0
10.0.123.0/24	20	Transit	10.0.123.1	10.0.1.1	0.0.0.0
10.0.2.2/32	20	Stub	10.0.123.2	10.0.2.2	0.0.0.0
10.0.3.3/32	20	Stub	10.0.123.3	10.0.3.3	0.0.0.0

```
Total Nets: 4
```

```
Intra Area: 4  Inter Area: 0  ASE: 0  NSSA: 0
```

```
[R1]
```

The command output shows that the Loopback0 routes learned from R2 and R3 still exist.

3.15 Configuring Default Route Advertisement

Configure a default route on R3 to access the external network (the loopback interface of R4 is used as an example).

```
[R3]ip route-static 0.0.0.0 0.0.0.0 202.96.34.4
```



```
[R3]ping 202.96.1.1
PING 202.96.1.1: 56 data bytes, press CTRL_C to break
  Reply from 202.96.1.1: bytes=56 Sequence=1 ttl=255 time=40 ms
  Reply from 202.96.1.1: bytes=56 Sequence=2 ttl=255 time=20 ms
  Reply from 202.96.1.1: bytes=56 Sequence=3 ttl=255 time=30 ms
  Reply from 202.96.1.1: bytes=56 Sequence=4 ttl=255 time=10 ms
  Reply from 202.96.1.1: bytes=56 Sequence=5 ttl=255 time=30 ms

--- 202.96.1.1 ping statistics ---
  5 packet(s) transmitted
  5 packet(s) received
  0.00% packet loss
  round-trip min/avg/max = 10/26/40 ms
```

[R3]

R3 can ping the loopback interface of R4, but R1 and R2 cannot ping the loopback interface of R4 because they have no route to R4.

```
<R1>ping 202.96.1.1
PING 202.96.1.1: 56 data bytes, press CTRL_C to break
  Request time out
  Request time out
  Request time out
  Request time out
  Request time out

--- 202.96.1.1 ping statistics ---
  5 packet(s) transmitted
  0 packet(s) received
  100.00% packet loss
```

<R1>

```
<R2>ping 202.96.1.1
PING 202.96.1.1: 56 data bytes, press CTRL_C to break
  Request time out
  Request time out
  Request time out
  Request time out
  Request time out

--- 202.96.1.1 ping statistics ---
  5 packet(s) transmitted
  0 packet(s) received
  100.00% packet loss
```

<R2>

Configure R3 to advertise default OSPF routes.

```
[R3]ospf 1
[R3-ospf-1]default-route-advertise always cost 10
```



Check the OSPF routing tables on R1 and R2. The command output shows a default OSPF route.

<R1>display ospf routing

OSPF Process 1 with Router ID 10.0.1.1
Routing Tables

Routing for Network

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.1.1/32	0	Stub	10.0.1.1	10.0.1.1	0.0.0.0
10.0.123.0/24	20	Transit	10.0.123.1	10.0.1.1	0.0.0.0
10.0.2.2/32	20	Stub	10.0.123.2	10.0.2.2	0.0.0.0
10.0.3.3/32	20	Stub	10.0.123.3	10.0.3.3	0.0.0.0

Routing for ASEs

Destination	Cost	Type	Tag	NextHop	AdvRouter
0.0.0.0/0	10	Type2	1	10.0.123.3	10.0.3.3

Total Nets: 5

Intra Area: 4 Inter Area: 0 ASE: 1 NSSA: 0

<R1>

<R2>display ospf routing

OSPF Process 1 with Router ID 10.0.2.2
Routing Tables

Routing for Network

Destination	Cost	Type	NextHop	AdvRouter	Area
10.0.2.2/32	0	Stub	10.0.2.2	10.0.2.2	0.0.0.0
10.0.123.0/24	1	Transit	10.0.123.2	10.0.2.2	0.0.0.0
10.0.1.1/32	1	Stub	10.0.123.1	10.0.1.1	0.0.0.0
10.0.3.3/32	1	Stub	10.0.123.3	10.0.3.3	0.0.0.0

Routing for ASEs

Destination	Cost	Type	Tag	NextHop	AdvRouter
0.0.0.0/0	10	Type2	1	10.0.123.3	10.0.3.3

Total Nets: 5

Intra Area: 4 Inter Area: 0 ASE: 1 NSSA: 0

<R2>

On R1 and R2, ping the IP address 202.96.1.1 of the loopback interface on R4. The ping is successful.

<R1>ping 202.96.1.1

PING 202.96.1.1: 56 data bytes, press CTRL_C to break

Reply from 202.96.1.1: bytes=56 Sequence=1 ttl=254 time=50 ms



```
Reply from 202.96.1.1: bytes=56 Sequence=2 ttl=254 time=70 ms
Reply from 202.96.1.1: bytes=56 Sequence=3 ttl=254 time=40 ms
Reply from 202.96.1.1: bytes=56 Sequence=4 ttl=254 time=60 ms
Reply from 202.96.1.1: bytes=56 Sequence=5 ttl=254 time=50 ms

--- 202.96.1.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 40/54/70 ms

<R1>

<R2>ping 202.96.1.1
 PING 202.96.1.1: 56 data bytes, press CTRL_C to break
  Reply from 202.96.1.1: bytes=56 Sequence=1 ttl=254 time=60 ms
  Reply from 202.96.1.1: bytes=56 Sequence=2 ttl=254 time=50 ms
  Reply from 202.96.1.1: bytes=56 Sequence=3 ttl=254 time=60 ms
  Reply from 202.96.1.1: bytes=56 Sequence=4 ttl=254 time=50 ms
  Reply from 202.96.1.1: bytes=56 Sequence=5 ttl=254 time=60 ms

--- 202.96.1.1 ping statistics ---
 5 packet(s) transmitted
 5 packet(s) received
 0.00% packet loss
 round-trip min/avg/max = 50/56/60 ms

<R2>
```

4 Verification

Verification has been performed during the lab.

5 Quiz

1. Analyze which interfaces can be configured as silent interfaces in actual networking scenarios.
2. (Multiple-choice answer) Which of the following states are stable during the establishment of OSPF neighbor relationships and adjacencies? (☐ BD ☐)
 - A. Exstart
 - B. Two-way
 - C. Exchange
 - D. Full
3. (Multiple-choice answer) In which of the following situations will an adjacency be established between routers? (☐ ABD ☐)
 - A. Two routers on a point-to-point link
 - B. DR and BDR on a broadcast network
 - C. DR Others on an NBMA network
 - D. BDR and DR Others on a broadcast network